

Introduction to Artificial Intelligence

Artificial Intelligence (AI) is a field of computer science focused on creating systems that can perform tasks that normally require human intelligence. These tasks include understanding language, recognizing images, solving problems, and making decisions.

Machine Learning

Machine Learning (ML) is a subset of AI where computers learn patterns from data instead of being explicitly programmed. Common types of machine learning include supervised learning, unsupervised learning, and reinforcement learning.

Supervised Learning

In supervised learning, the model learns from labeled data. Examples include predicting house prices using regression or classifying emails as spam or not spam using classification algorithms.

Unsupervised Learning

Unsupervised learning works with unlabeled data. The model tries to find hidden patterns or groupings. Clustering algorithms such as K-Means are popular examples.

Natural Language Processing

Natural Language Processing (NLP) enables computers to understand and generate human language. Applications include chatbots, sentiment analysis, translation systems, and question-answering systems.

Retrieval Augmented Generation

Retrieval Augmented Generation (RAG) is a technique where a system retrieves relevant information from a document database and uses a language model to generate answers based on that information.